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	First we will fully develop the crosslinking experiments using a variety of crosslinking reagents that are known to react rapidly under the acidic conditions of tetragonal lysozyme crystal growth. The crosslinking reaction will be used to study aggregates in solutions that lead to crystal growth of lysozyme and insulin. We will explore the temperature and salt dependence of the aggregate distributions. From the temperature dependence we will determine the thermodynamic parameters which will be used to understand the intermolecular forces which predominate under both crystallizing and precipitating conditions. We will apply simplex optimization techniques to protein crystallization by optimizing the conditions for crystal growth of the well-characterized systems of insulin and lysozyme. This approach will be applied to the crystallization of human serum transferrin, a more challenging crystallization problem. If simplex optimization leads to diffraction quality crystals of transferrin we plan to solve it’s monoferric structure using molecular replacement techniques. There are three primary objectives of our work:		

Task Description:	1) Fully characterize metastable solutions of proteins under solution conditions which lead to precipitate and crystallization. The presence and distribution of aggregates in supersaturated solutions will be investigated by chemical cross-linking followed by SDS-PAGE, size-exclusion chromatography and capillary electrophoresis. The role of precipitating agents in aggregate formation will be studied using different inorganic salts and polyethylene glycol. The time dependence of prenucleation clusters will be analyzed using turbidity analysis. 2) Determine if aggregation rates (aggregates formed as a function of time) are lowered in microgravity by carrying out chemical cross-linking experiments on Space Shuttle flights in the Fluid Processing Apparatus. If a decrease in the aggregation rate is found, then fewer, and possibly smaller, aggregates would be present during the growth stage. With fewer aggregates present we propose that smaller, more ordered, aggregates would be the favored growth units. It is possible that this, and a reduced transport of impurities or nonspecific aggregates to the surface, could be the explanation for the higher quality protein crystals seen in microgravity. 3) Develop simplex optimization techniques for protein crystal growth to be used by investigators doing microgravity crystallization experiments on the International Space Station. This approach requires fewer experiments than both the single-factor-at-a-time approach or factorial designs and will conserve material on Space Station. In addition, it will allow optimization of conditions for microgravity which are probably different than the optimum conditions for crystal growth on Earth. With the growing need for new medications to fight disease structure-based drug design is becoming more important. The first step in the process to design a new drug involves crystallization of an important biological macromolecule. Our work looks at the crystallization process in an attempt to further understand it and design new methods for crystal growth. This will have implications not only in drug design but also in the understanding of many biological mechanisms. The long-term objectives of the proposed research are to further the understanding of the fundamental processes of protein nucleation and crystal growth both on the ground and in microgravity. With the recent results from NASA's microgravity crystallization experiments we now plan to answer the following questions using our experimental techniques: 1) Does gravity affect the formation of aggregates? 2) Does gravity affect the size distribution and/or average size of the aggregates? 3) Do the aggregates play a different role in the crystal growth process in microgravity than on Earth? 4) Does gravity affect the transport of the aggregates to the crystal surface? Answers to these important questions will allow crystallographers to design crystallization methods that control aggregate formation thereby increasing the quality of crystals obtained on the ground and in microgravity. This will lead to better crystal structures which has implications in the fields of drug design, protein engineering, and many areas of biomedical research.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This work investigates the nucleation and growth of protein crystals, which are used to determine protein structures, and eventually leads to the development of new drugs.
	Ground based research is being conducted in the Department of Chemistry at Eastern Kentucky University (EKU) in Richmond, KY. We have determined the conditions of purity and supersaturation required for the microgravity flight experiment, XLINK, which uses chemical crosslinking to study differences in aggreage distributions on Earth and in Space. The hardware for the flight experiment was purchased from BioServ Space Technologies in Boulder, CO and is called a Group Activation Pack or GAP which holds 8 Fluid Processing Apparati (FPAs). The hardware is being tested at EKU in a series of experiments to ensure proper mixing and working conditions. The crosslinking experiment is one which uses glutaraldehye to chemically crosslink protein aggregates in supersaturated protein solutions. In microgravity we will vary amount of precipitant used and the supersaturation level. Because the crosslinked aggregates are stable once formed all characterization will be done in the lab after the samples return to Earth. For microgravity, this approach has many advantages over more expensive and complicated light scattering techniques. First, these experiments are simple to perform and require mixing of solutions that can be done using commercially available flight hardware that is already built and fully tested. Second, all analysis of data is done off line in the laboratory so that no recording, monitoring and downlinking of data is necessary. In support of the XLINK investigation we are providing the following: 1. Provide support to Bioserve necessary to conduct the flight experiment. 2. Furnish any required data to Bioserve as well as necessary reports required by NASA. 3. Prepare and provide experiment sample solutions for flight. 4. Load experiment sample solutions into flight hardware containers. 5. Conduct ground-based research to support the flight investigation including determining proper mixing and experiment conditions. 6. Provide science requirements for this investigation to SPD. 7. Support mission operations, launch and landing preparations, and post-flight experiment recovery. 8. Support and/or conduct training of flight crew. The PI visited Bioserv in May, 2004. Mixing. Since this experiment relies on adequate mixing of crosslinking reagent throughout a superaturated protein solution a device was developed which allowed us to visualize the mixing of different density solutions in microgravity. In order to facilitate this chaotic flow, and thus produce a homogeneous mixture, a stir vane will be placed on the bottom and side of a cubic container. The device was tested on the KC-135 in August, 2003, the first four parabolas were used to observe static motion representing molecular diffusion as a control. During subsequent parabolas, the spin vanes were activated in sequences ranging from single plane, both horizontal and vertical, and finally in multi-plane mode varying the mixing time. To catalyze mixing, the stir vanes were turned on and off in alternating clockwise and counterclockwise rotations. The effect on mixing was visually analyzed and modifications were made in flight to aid in complete mixing. In order to properly visualize the effects of mixing and create a situation in which the environment will be “reset” through the gravity phase of the experiment, low-density polystyrene beads and high-density Teflon® beads were placed in the water filled container. Upon entering the microgravity phase of the experiment, water agitation dispersed the particles throughout the solution. Once gravity was regained, the particles returned to their original location in the solution based on density thus allowing for multiple runs without complicated separations. Dispersion analysis is currently underway using the digital videotaped results of the experiment. Image processing software is being

Task Progress:	used to divide the image into a square grid of equal subdivisions. These quantities will be examined for uniformity and statistical variances determined. Since the images were taken with a video camera, time analysis of dispersion from t = 0 to t = x, where x is the time of the zero gravity interval, will be determined and a 100 point turbulent fluid dispersion model will be generated. In the first series of microgravity simulations, static motion was observed in order to develop a control. In sequential series, the spin-vanes were operated in different capacities to produce the respective mixing environments. The sequence will be the same as performed on the ground unless in-flight visual inspection of the mixing shows a need to alter the mixing procedure to produce better results. Attempts will be made to model both the static motion representing molecular diffusion and the turbulent mixing scenarios using different turbulent flow models. Temperature. Storage Temperature: Currently, we are considering flying XLINK using The Low Temperature, Low-Energy Carrier (LoTEC) which is designed to carry temperature sensitive materials back and forth to the International Space Station (ISS) without using external sources of power. Since LoTEC is not powered the temperature of storage will vary depending on the cabin temperature (if stored in the middeck). To determine the affect of temperature fluctuations on our solutions we are testing the long term storage at different temperatures. We are storing loaded GAPs at temperatures ranging from 25oC to 40oC. In addition we are growing crystals of lysozyme and looking for phase changes over the same temperature range. Reaction Temperature: Since the microgravity experiment will be run at station ambient conditions we need to determine if the aggregate equilibrium are temperature sensitive. Given that solubility changes with temperature it is likely that they are affected and careful controls will be necessary. Analysis at different temperature resulted in a temperature effect that is in the reverse of the solubility effect (more aggregates at higher temperature). This suggests to us that the increases in the temperature increases the crosslinking reaction rate. Therefore, we cannot study the temperature dependence of aggregation with this method. It also underscores the temperature sensitivity of the microgravity reaction. We will conduct temperature sensitivity studies on the pre- and post-reaction solutions. Quantitation. We will compare the number and types of aggregates obtained in microgravity to those obtained on the ground. Ground based results will be repeated with the same solutions that are used in the microgravity experiments. This will allow us to determine if the crosslinks remain stable while we wait for the experiments to return to Earth. Our experience is that the crosslinks are stable and there is little change with time if the reaction is terminated using hydrazine or some other amine. We will separate the aggregates by size exclusion chromatography with UV detection and SDS-CE, which we have found to provide better quantification for detection than SDS-PAGE with silver stain. Solution Conditions. For the microgravity experiment we are suggesting that the concentration of the precipitating agent, NaCl, be varied from 0 to 15%. This covers the entire range of supersaturations from undersaturated (0% NaCl), metastable (3-6% NaCl) and precipitation regime (12% NaCl). The molar ratio of glutaraldehyde to protein will be varied from 100 to 1000. This represents a low and high ratio of crosslinker to protein, respectively. If the ratio is too high precipitation of the protein at high salt concentrations has been known to occur. Conversely, if this ratio is too low no crosslinks are obtained. Stability. We are conducting stability studies on pre- and post-reaction solutions. This includes the affect of separation methods for the quantitation studies mentioned above. Two of the methods mentioned use denaturing conditions, which could possibly affect the crosslinks. If necessary we may stabilize the Schiff base with a reducing agent. Finally, we will investigate the need to degas the solutions. X-link Fluids. We have developed a list of fluids and their concentrations. This list has been sent to BioServ Space Technologies and then forwarded on to JSC safety. X-link Timeline. Working with BioServe Space Technologies we have developed a timeline to be used in crew training procedures. X-link Video: Working with BioServe Space Technologies we have developed a video imaging layout. Imaging is important in order for us to determine if an when precipitation of the solution occurs. Precipitation can occur during any stage of this multistage reactions: 1) upon intiation of supersaturation 2) during crosslinking and/or 3) upon termination of crosslinking. Analysis of results is dependent on this information.
Bibliography Type:	Description: (Last Updated: 02/10/2006)
Articles in Peer-reviewed Journals	Hall, C.L., Clemens, J.R., Brown, A.M., Hu, H., Wilson, L.J. "Even Unit Addition to Protein Aggregates" submitted to J. Crystal Growth , Jun-2004
Presentation	Clemens, J., Brown, A., Hall, C., Lanham, K., Myers, G., Norwitz, S., Penn, W., Harvey, C., Wilson, L., "Multiplanar Microgravity Mixing" Undergraduate Presentation Showcase, Scholars Weekend, EKU Apr-2004
Presentation	Clemens, J., Brown, A., Hall, C., Lanham, K., Myers, G., Norwitz, S., Penn, W., Harvey, C., Wilson, L "Analysis of multi-planar mixing in a microgravity environment" 9th Annual EPSCoR Conference, Advances in Research Through EPSCoR May-2004